

Tableau Prep and TabPy Integration Report

This report demonstrates the integration of Tableau Prep with Python using TabPy. The task involved loading an employee CSV dataset, applying a Python script through a Script Step, filtering IT department employees, calculating the average salary, and identifying the oldest employee.

Steps Performed

Step	Description
1	Connected Tableau Prep to employee_tableau_prep_python_sample.csv
2	Started TabPy server on localhost:9004
3	Added a Script Step in Tableau Prep
4	Connected the Script Step to TabPy
5	Executed a Python script to process employee data
6	Generated the transformed output and exported as CSV

Python Script Used

```
from pandas import DataFrame

def run(df):

    it_employees = df[df['Department'] == 'IT'].copy()

    avg_salary = df['Salary'].mean()

    oldest_employee = df.loc[df['Age'].idxmax()]

    oldest_name = oldest_employee['Employee_Name']
    oldest_department = oldest_employee['Department']

    it_employees['Average_Salary_All_Employees'] = avg_salary
    it_employees['Oldest_Employee_Name'] = oldest_name
    it_employees['Oldest_Employee_Department'] = oldest_department

    return it_employees
```

Final Output

Employee_ID	Employee_Name	Department	Average Salary	Oldest Employee	Oldest Department
E001	Anna Weber	IT	57,100	Helga Braun	HR
E003	Sara Hoffmann	IT	57,100	Helga Braun	HR
E006	Omar Yilmaz	IT	57,100	Helga Braun	HR
E009	Maya Becker	IT	57,100	Helga Braun	HR
E012	Tim Schneider	IT	57,100	Helga Braun	HR

The integration between Tableau Prep and TabPy was successfully established. The Python script executed correctly and produced the required transformed dataset. The final output included only IT employees along with calculated salary and oldest employee details.